Ch 5.1.1-2: Leave One Out Cross-validation

Lecture 10 - CMSE 381

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Announcements

Last time:

- Exam

Announcements:

- Fourth homework due next monday
- Office hours
- Drops
- Grade conversion

Percent	Convert
≥ 90%	4.0
≥ 85%	3.5
≥ 80%	3
≥ 75%	2.5
≥ 70%	2
≥ 65%	1.5
≥ 60%	1
< 60%	0

Covered in this lecture

- Validation Set testing
- LOO CV
- Outliers
- Leverage statistic

Section 1

Validation set

What's the problem?

- How well is my ML method doing? *Model Assessment* → MSE (regression) % incorrect (classification)
- Which method is best for our data?
- How many features should I use? Which ones? *Model selection*
- What is the uncertainty in the learned parameters?

How do I choose hyper parameters?

Ex) degree of polynomial

Input to model
not learned by model

Training Error vs Testing Error

Training Error

Calculated by applying stats learning method to specific set of observations used in the training of the model.

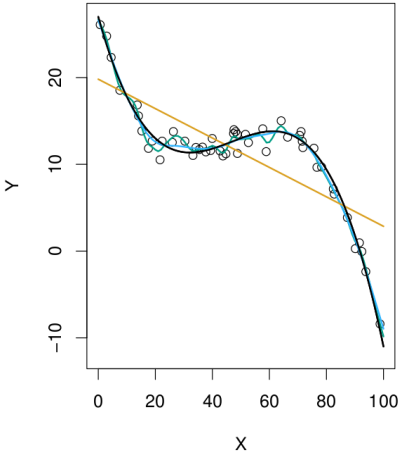
Massive underestimate for testing error

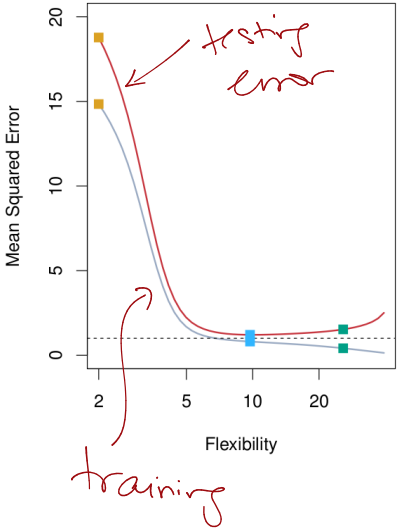
Overfitting

Testing Error

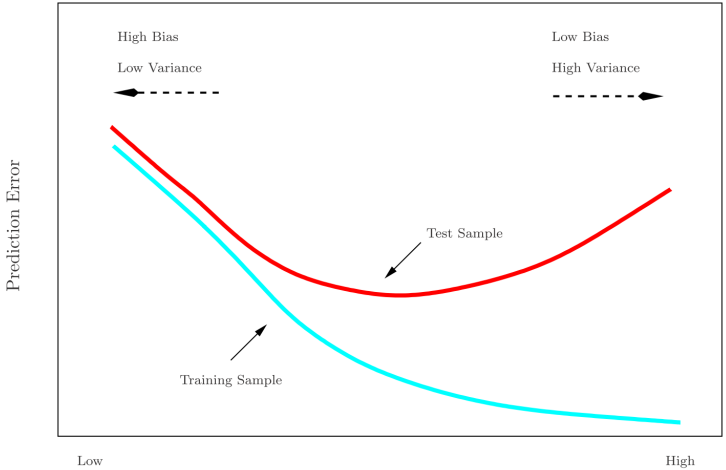
Avg error that results from using a statistical learning method to predict a response on a new observation, one not used in the model training.

Throw-back ~~Wednesday~~ Monday





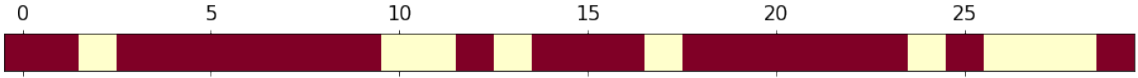
Model tradeoffs



, hyper parameter controlled

Validation set approach

Index of data points



- Divide randomly into two parts:
  - Training set → red
  - Validation/Hold-out/Testing set → Yellow pts
- Fit model on training set
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

fix percentage of data

Ex) 30% is testing

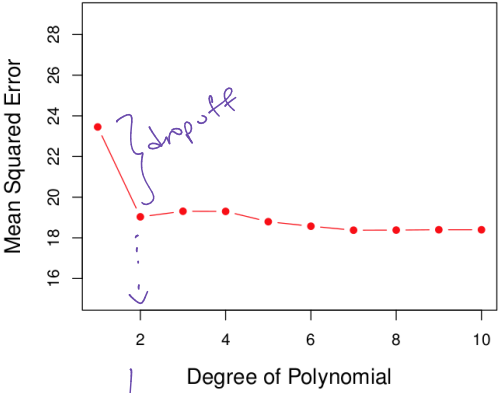
No "data snooping"

Do not use testing data for training

on the test set

Coding example in jupyter notebook

Example with the `auto` data



want $p=2$

Predicting `mpg` using `horsepower`:

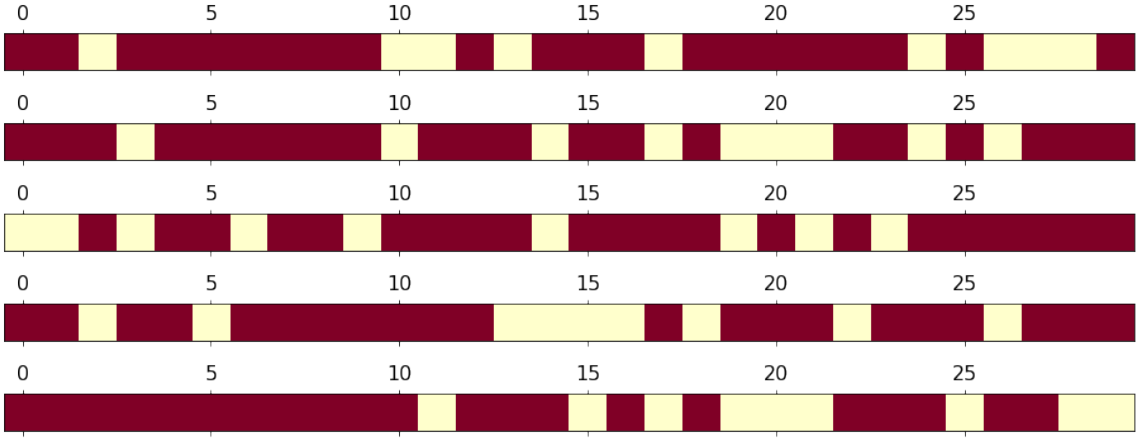
$$\texttt{mpg} = \beta_0 + \beta_1 \texttt{hp} + \beta_2 \texttt{hp}^2 + \cdots + \beta_p \texttt{hp}^p$$

Test error

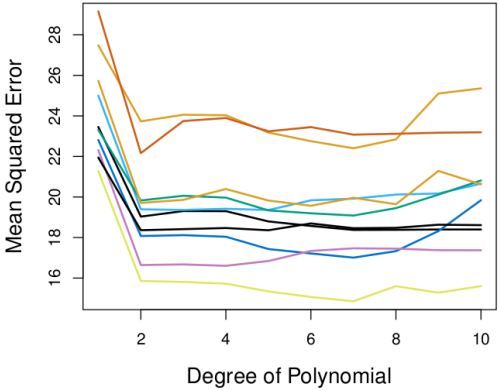
using validation set

Each using the same validation set

Rinse and repeat



Again example with auto data

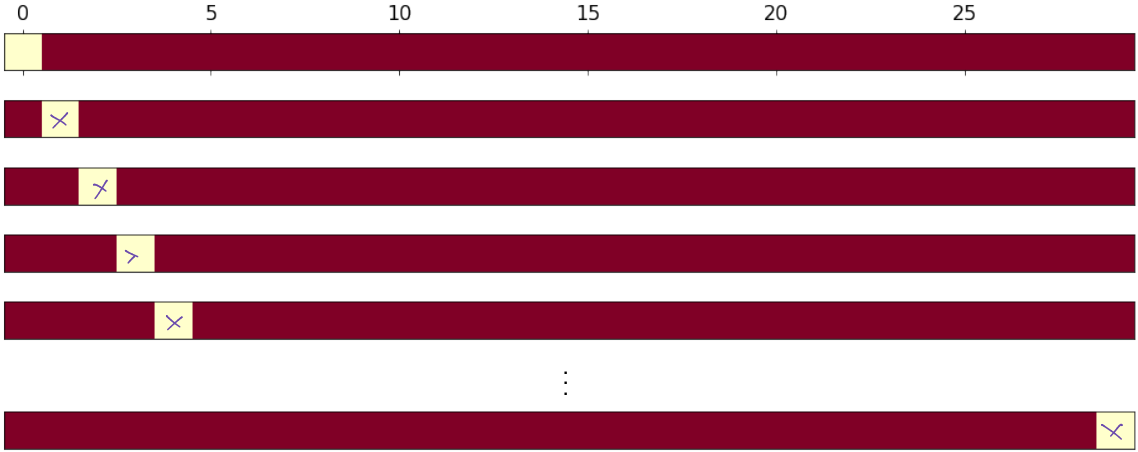


- Each line: using same train/test split for all p
- While the shape is similar
- What is the true testing error?
  Can't tell b/c so variable
- Stats models trained on less data (held out some for testing) tend to do worse

Section 2

Leave-One-Out Cross-Validation (LOOCV)

The idea



The idea in mathy words

- Remove (x_1, y_1) for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1)^2$

- Remove (x_2, y_2) for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2)^2$

- Rinse and repeat

Return the score:

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

Do the LOOCV coding section

LOOCV Pros and Cons

Advantages:

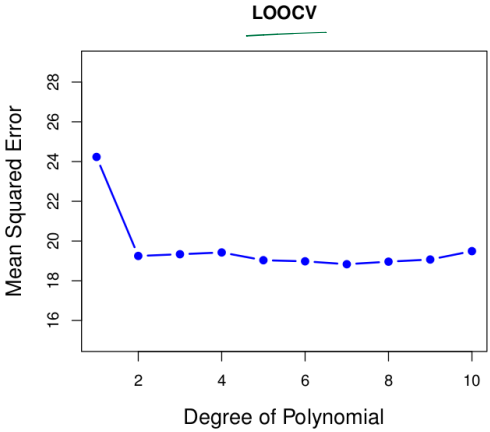
- Use almost all data every time.
- Stable results: No randomness! always get same answer on a fixed data set.
- Evaluation always performed on more testing data

Disadvantages:

Computationally expensive

fit model n times

Again example with auto data



Section 3

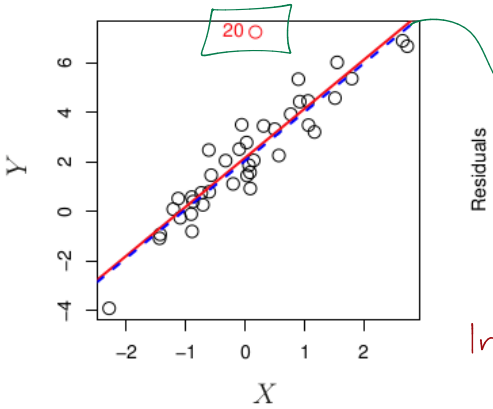
The one time you can cheat (by not computing every model fit)

Linear
Regression

Outliers

An *outlier* is a point for which y_i is far from the value predicted by the model.

↳ $|y_i - \hat{y}_i|$ is large



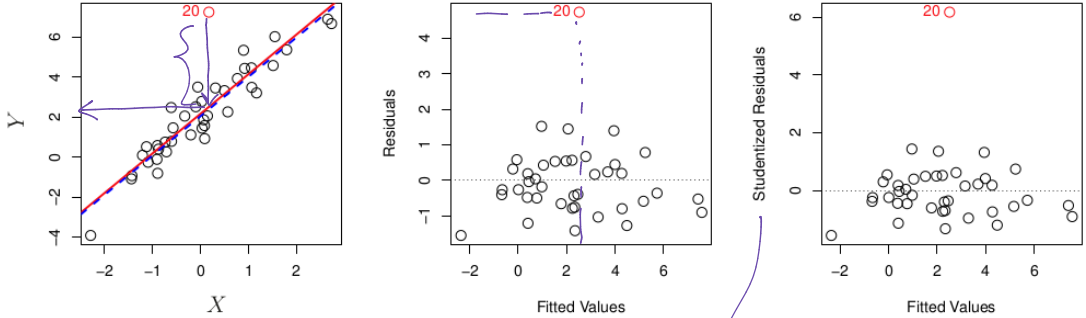
Red solid
Regression learned w/ all data

Blue dashed
Regression learned w/o pt 20

In this case, not much effect on the learned model.

Residuals

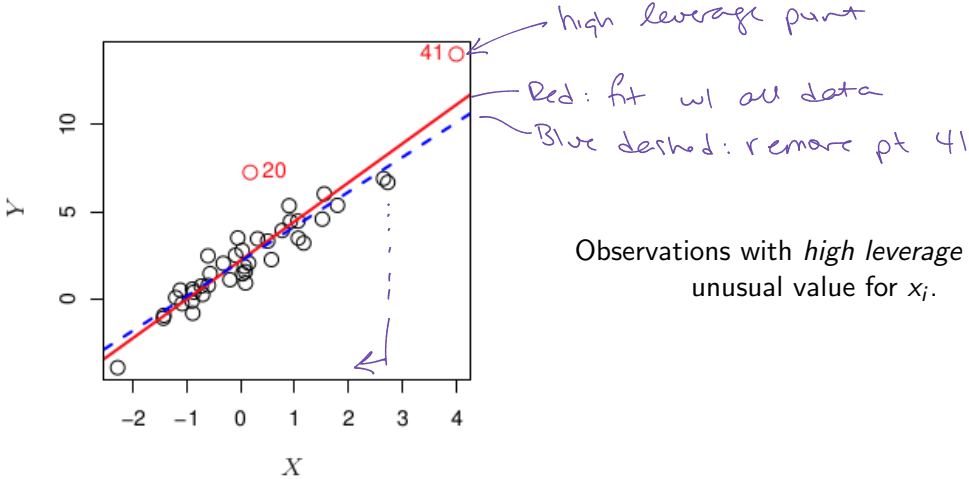
$\hat{y}_i - y_i$



Common choice:

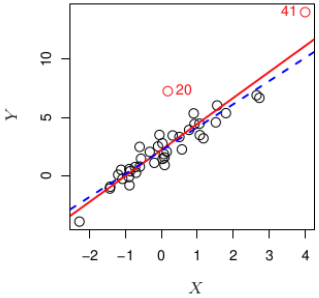
$\geq 3 \rightsquigarrow$ outlier

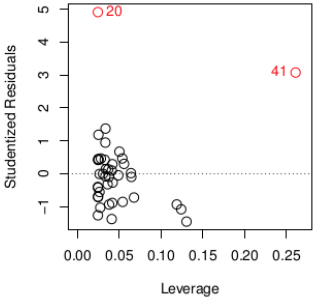
High Leverage



Observations with *high leverage* have an unusual value for x_i.

Leverage statistic





Version for $p = 1$

$$h_i = \frac{1}{n} + \frac{(x_i - \overline{x})^2}{\sum_{j=1}^{n}(x_j - \overline{x})^2}$$

Computer can calculate for $p > 1$

Leverage statistic properties

$$h_i = \frac{1}{n} + \frac{(x_i - \overline{x})^2}{\sum_{j=1}^{n}(x_j - \overline{x})^2}$$

- Always increases with distance of x_i from $\overline{x}$
- Always in $[1/n, 1]$
- Avg leverage is $\frac{p+1}{n}$
- Prev page: 41 both high leverage and outlier

  20 only outlier, so little effect on the regression

Speeding up LOOCV

Warning: This only works for least squares linear or polynomial regression.

original definition

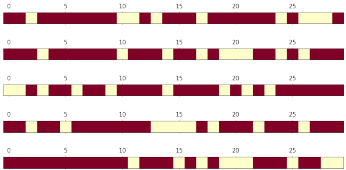
prediction when fit using all n data points

$$\underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathrm{MSE}_i}_{} = CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2$$

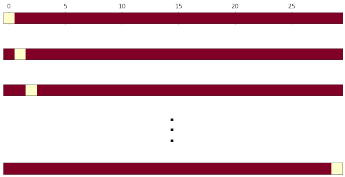
only works with regression

TL;DR

Validation set



LOO-CV



LOO-CV Score

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

Cheap trick for regression

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2$$

Next time

14	M	Oct 3	Leave one out CV	5.1.1, 5.1.2	
15	W	Oct 5	k-fold CV	5.1.3	
16	F	Oct 7	More k-fold CV	5.1.4	
17	M	Oct 10	CV for classification	5.1.5	HW #4 Due
18	W	Oct 12	Resampling methods: Bootstrap	5.2	
19	F	Oct 14	Subset selection	6.1	
20	M	Oct 17	Shrinkage: Ridge	6.2.1	HW #5 Due
21	W	Oct 19	Shrinkage: Lasso	6.2.2	
22	F	Oct 21	Dimension Reduction	6.3	
	M	Oct 24	No class - Fall break		
21	W	Oct 26	More dimension reduction; High dimensions	6.4	
22	F	Oct 28	Polynomial & Step Functions.	7.1,7.2	HW #6 Due
23	M	Oct 31	Review		
24	W	Nov 2	***Midterm #2***		